WORKED SOLUTION

STEP II 2012, Question 8

Pure • Non-Linear Recurrence Relations

STEP II 2012, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q8

The key idea

The recurrence $u_{n+1}u_{n-1} = u_n^2 - q^2$ is non-linear, but it holds for *every* n, so writing it down for two consecutive values of n and subtracting kills the awkward constant q^2. What survives says that the ratio $(u_{n+1} + u_{n-1})/u_n$ never changes, and that turns the sequence into a *linear* recurrence with constant coefficient p. The opening inequality is exactly the statement $p > 2$, which is what drives the growth.

Solution

The inequality

Putting everything over the common denominator $\alpha\beta$,

$$\frac{\alpha^2 + \beta^2 - q^2}{\alpha\beta} - 2 = \frac{\alpha^2 - 2\alpha\beta + \beta^2 - q^2}{\alpha\beta} = \frac{(\beta - \alpha)^2 - q^2}{\alpha\beta}.$$

Since $\beta - \alpha > q > 0$, squaring (both sides positive) gives $(\beta - \alpha)^2 > q^2$, so the numerator is positive. The denominator $\alpha\beta$ is positive because $\alpha, \beta > 0$. Hence the expression is positive, as required.

The first identity

Multiply the recurrence through by u_{n-1}: for every $n \geq 1$,

$$u_{n+1}u_{n-1} = u_n^2 - q^2.$$

Replacing n by $n + 1$ (still allowed, since $n + 1 \geq 1$),

$$u_{n+2}u_n = u_{n+1}^2 - q^2.$$

Subtracting the first equation from the second removes q^2:

$$u_n u_{n+2} - u_{n-1}u_{n+1} = u_{n+1}^2 - u_n^2,$$

and rearranging,

$$u_n^2 + u_n u_{n+2} = u_{n+1}^2 + u_{n-1}u_{n+1}, \qquad \text{that is,} \qquad u_n(u_n + u_{n+2}) = u_{n+1}(u_{n-1} + u_{n+1})$$

for all $n \geq 1$, which is the required result.

The linear recurrence

> **Ask yourself** The identity just proved has four different terms in it. How does it give a recurrence with a single constant p?

Divide it by $u_n u_{n+1}$, which is allowed because no term of the sequence is zero:

$$\frac{u_{n+2} + u_n}{u_{n+1}} = \frac{u_{n+1} + u_{n-1}}{u_n} \qquad (n \geq 1).$$

So the quantity $v_n = (u_{n+1} + u_{n-1})/u_n$ satisfies $v_{n+1} = v_n$ for all $n \geq 1$: it is constant. Call its common value p; then $u_{n+1} + u_{n-1} = pu_n$, i.e.

$$u_{n+1} - pu_n + u_{n-1} = 0 \qquad (n \geq 1).$$

To find p, use $n = 1$. From the recurrence, $u_2 = (\beta^2 - q^2)/\alpha$, so

$$p = \frac{u_2 + u_0}{u_1} = \frac{1}{\beta}\left(\frac{\beta^2 - q^2}{\alpha} + \alpha\right) = \frac{\alpha^2 + \beta^2 - q^2}{\alpha\beta}.$$

This is the expression from the first part, so when $\beta - \alpha > q$ we already know that $p > 2$.

The sequence is increasing when $\beta > \alpha + q$

Let $d_n = u_{n+1} - u_n$. Using the linear recurrence in the form $u_{n+1} = pu_n - u_{n-1}$,

$$d_n - d_{n-1} = u_{n+1} - 2u_n + u_{n-1} = (p-2)u_n.$$

> **Ask yourself** We know $p - 2 > 0$, but why should $(p-2)u_n$ be positive? We don't yet know that $u_n > 0$.

Positivity and growth have to be proved together, by induction. Claim: for every $n \geq 1$, $u_n > 0$ and $d_{n-1} > 0$.

For $n = 1$: $u_1 = \beta > 0$ and $d_0 = \beta - \alpha > q > 0$.

Suppose $u_n > 0$ and $d_{n-1} > 0$ for some $n \geq 1$. Then, since $p > 2$,

$$d_n = d_{n-1} + (p-2)u_n > 0, \qquad u_{n+1} = u_n + d_n > 0,$$

which is the claim for $n + 1$. By induction, $d_n = u_{n+1} - u_n > 0$ for every $n \geq 0$, so the sequence is strictly increasing. (In fact the differences themselves increase, so the growth is faster than linear. Since every term is positive, the condition that no term is zero holds automatically in this case.)

The case $\beta = \alpha + q$

Now $(\beta - \alpha)^2 - q^2 = 0$, so $p = 2$ exactly and $d_n - d_{n-1} = 0$. The differences are all equal to $d_0 = \beta - \alpha = q$, so the sequence is the arithmetic progression

$$u_n = \alpha + nq.$$

It is still strictly increasing, since $q > 0$, but now only linearly. Direct check: $(\alpha + (n+1)q)(\alpha + (n-1)q) = (\alpha + nq)^2 - q^2$, so this sequence does satisfy the original recurrence.

Checks

- Take $\alpha = 1$, $\beta = 4$, $q = 2$ (so $\beta > \alpha + q$). The recurrence gives $1, 4, 12, 35, \frac{407}{4}, \ldots$, and $p = \frac{1+16-4}{4} = \frac{13}{4}$; indeed $\frac{13}{4} \cdot 4 - 1 = 12$ and $\frac{13}{4} \cdot 12 - 4 = 35$.
- With $\alpha = 1$, $q = 2$, $\beta = 3$ the terms are $1, 3, 5, 7, \ldots$, the arithmetic progression of the last part.
- We checked both identities in exact rational arithmetic for hundreds of random α, β, q, and checked that the sequence increases whenever $\beta > \alpha + q$.

Common slips

- Concluding $d_n > d_{n-1}$ from $p > 2$ without first knowing that $u_n > 0$: the two facts must go into the same induction.
- Dividing by $u_n u_{n+1}$ without remarking that it is non-zero.
- Forgetting that the subtraction step uses the recurrence at index $n + 1$, which needs $n + 1 \geq 1$.

Beyond STEP

A non-linear recurrence that secretly satisfies a linear one is said to have a hidden *invariant*: here $(u_{n+1} + u_{n-1})/u_n$, or equivalently $u_n^2 - u_{n+1}u_{n-1} = q^2$, which is a discrete analogue of a conserved quantity in mechanics. The linear recurrence $u_{n+1} - pu_n + u_{n-1} = 0$ is solved by $u_n = A\lambda^n + B\lambda^{-n}$ where $\lambda + \lambda^{-1} = p$; this is real and exponentially growing when $p > 2$, and becomes $A + Bn$ in the repeated-root case $p = 2$, exactly the dichotomy above. Siklos, *Advanced Problems in Mathematics*, has more STEP-style recurrences, and the parallel with differential equations is developed in Boyce and DiPrima, *Elementary Differential Equations*.

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